

No. of Functional Features Included in the Solution

Date	06 JULY 2026
Team ID	SWTID-2026-8308
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Login & Registration	Allows farmers and administrators to securely register and log in.	Authentication	Done	10%
2	Crop Recommendation	Recommends the best crop based on soil type, weather, and season using AI/ML.	Recommendation Engine	Done	20%
3	Weather Forecast Integration	Displays real-time weather updates to help farmers plan agricultural activities.	Weather Module	Done	15%
4	Soil Analysis	Analyzes soil parameters such as pH, moisture, and nutrients for better crop selection.	Soil Analysis Module	Done	15%
5	Fertilizer Recommendation	Suggests suitable fertilizers based on crop and soil condition.	Fertilizer Recommendation	Done	10%
6	Irrigation Management	Recommends optimal irrigation schedules to reduce water wastage.	Irrigation Module	Done	10%

7	Yield Prediction	Predicts expected crop yield using historical and environmental data.	Prediction Engine	In Progress	10%
8	Dashboard & Reports	Provides visual reports, analytics, and farming insights through graphs and dashboards.	Dashboard & Reporting	Done	10%

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	User Authentication, Crop Recommendation, Soil Analysis, Weather Forecast, Fertilizer Recommendation
Additional / Nice-to-Have Features	Irrigation Management, Yield Prediction, Dashboard & Reports
Features Tested & Verified	7 Features Successfully Tested and Verified

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Farmer-friendly interface, Dashboard, Reports	Login Page, Crop Recommendation Screen, Weather Dashboard
2	Backend / Logic	AI-based crop recommendation, Yield prediction, Fertilizer recommendation	Crop Recommendation Engine, Irrigation Logic, Yield Prediction Module
3	Database / Storage	Stores user details, crop data, soil records, weather history	Farmer Database, Crop Database, Soil & Weather Data Storage
4	API / Integration	External API integration for weather and agricultural information	Weather API, Soil Data API, SMS/Email Notification API
5	Security / Authentication	Secure user access and data protection	User Login, Registration, Password Encryption, Role-Based Access